Expense Management API

Business Requirements Document · Version 1.0 · 2026-10-21

1. Purpose

Provide a simple REST API that lets users record their expenses, organize them by category, and see how much they spend. The API will back a web or mobile expense management app.

2. Scope

In scope: user accounts, expense CRUD, categories, basic filtering, and a spending summary.
Out of scope (v1): receipt uploads, multi-currency conversion, approval workflows, bank integrations, team/shared budgets, notifications.

3. Users

Role	Description
User	A registered person who manages only their own expenses and categories.

4. Business Requirements

ID	Requirement
BR-1	A person can register and log in with email and password.
BR-2	A user can create, view, update, and delete their own expenses.
BR-3	Each expense has an amount, date, category, and optional description.
BR-4	A user can create, view, rename, and delete categories (e.g. Food, Transport, Rent).
BR-5	A user can list expenses and filter by date range and category.
BR-6	A user can view total spending for a period, broken down by category.
BR-7	A user can never see or change another user's data.

5. Data

Entity	Fields
User	id, email, password, name
Category	id, name
Expense	id, amount, currency, date, categoryId, description

6. Business Rules

- Amount must be greater than zero.
- Date is required and cannot be in the future.
- Each expense must belong to one category.
- A category that still has expenses cannot be deleted.
- Category names are unique per user.
- Email addresses are unique.
- A single currency per user (default USD) is used in v1.

7. API Capabilities (high level)

Area	Operations
Auth	Register, login
Expenses	Create, list (with filters), get one, update, delete
Categories	Create, list, rename, delete
Summary	Get totals by category for a date range

8. Non-Functional Requirements

- JSON over HTTPS; all endpoints except register/login require authentication.
- Passwords stored hashed, never returned.
- Standard HTTP status codes and clear error messages for invalid input.
- List endpoints support pagination.
- Typical requests respond in under 500 ms.

9. Success Criteria

- A user can register, add an expense, and see it in their summary.
- All business rules in section 6 are enforced and covered by tests.
- No user can access another user's data.

10. Assumptions

- Single currency per user; no currency conversion.
- Authentication is simple token-based; no social login.